Q2指南

> 这道大题主要考察Indexing, Preprocessing, BM25和Boolean Operators的相关概念知识，概念比较多，往届题复现率低，计算题考察TF-IDF构建向量以及计算cosine sim

Indexing

为什么inverted index比term-document incidence matrix更适合代表文档库 (24Q2.a(i))

An term-document incidence matrix does not scale well for larger document collections.

- Space complecxity: Inverted index only store non-zero values
- Scalability: Only need to create a new posting-list, rather than create a new row in the matrix
- Retrieval performance: Inverted index shows higher efficiency in both single query and boolean queries.

Preprocessing

Stopword Removal → Normalzation → Stemming

1. **Stopword removal** is the process of filtering out common occurring term that **does not add to the meaning**
2. **Normalzation** is the process that normalise the word, like lowercase capital letter or punctuation removal.
3. **Stemming/Lemmatization**: Process of reducing words to their root form/NLP technique that converts words to their base/dictionary form (lemma)

什么是Stopward removal, 怎么帮助IR process (24Q2.b(i))

- A **stopword** is a commonly occurring term that appears in so many documents that it **does not add to the meaning** of the document, and appear in most English texts. (e.g. *the, and, of, etc.*)

Zipf's Law和Stopword removal有什么联系 (24Q2.b(ii))

- If we rank terms from the most commonly used to the least commonly-used in a large text collection, the second term is approximately ½ as common as the first, the third is approximately ⅓ as common as the first, etc.
- We use this to identify what terms count as stopwords in various languages.

Stemming and Lemmatisation定义，Adv & Disadv (24Q2.b(iii))

Technique	Definition	Advantages	Disadvantages	Example
Stemming	Process of reducing words to their root form	Fast processing	May overstem (cut too much)	"computing", "computer" → "comput"

Technique	Definition	Advantages	Disadvantages	Example
Lemmatization	NLP technique that converts words to their base/dictionary form (lemma)	More effective	Slower processing	"running" → "run"

Boolean Operators

query operators如何工作/影响文档数量，展示Set Theory中对应的运算符 (23Q2.a)

Operator	Purpose	Effect	Set Notation	Example
AND	**Narrows** search by requiring ALL search terms	Returns only matching all terms	$A \cap B$	cats AND dogs
OR	**Broadens** search by including ANY of the terms	Returns matches for any term	$A \cup B$	cats OR dogs
NOT	**Excludes specific** terms from results	Removes unwanted matches	$A \setminus B$	cats NOT dogs

Two ways in which the process of running Boolean queries can be optimised (23Q2.b)

1. **Dynamic Postings List Ordering**

- Approach: Process terms in increasing order of document frequency (shortest lists first).

2. **Skip Pointers** in Postings Lists

- Approach: Add skip pointers to postings lists to jump over non-matching document IDs.

BM25

BM25三个principle

- Term Frequency (**TF**): A way of ranking how important term is in a document collection.
- Inverse Document Frequency (**IDF**): Count of the number of times the term appears in the document.

- Document length: BM25 normalises the length of document in calculations.

TF IDF

$$sim_{BM25}(d_j,q) \sim \sum_{k_i \in d_j \wedge k_i \in q} \frac{f_{i,j} \times (1+k)}{f_{i,j} + k\left((1-b) + \frac{b \times len(d_j)}{avg\_doclen}\right)} \times log\left(\frac{N\text{-}n_i + 0.5}{n_i + 0.5}\right)$$

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